Ecotoxicology and Environmental Safety 257 (2023) 114960

Contents lists available at ScienceDirect

Ecotoxicology and Environmental Safety

journal homepage: www.elsevier.com/locate/ecoenv

VAR-tree model based spatio-temporal characterization and prediction of O_3 concentration in China

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ARTICLE INFO

Editor: Richard Handy

Keywords:
O_3
VAR
Tree model
XGBoost
Spatio-temporal characteristics

ABSTRACT

Ozone (O_3) pollution in the atmosphere is getting worse in many cities. In order to improve the accuracy of O_3 prediction and obtain the spatial distribution of O_3 concentration over a continuous period of time, this paper proposes a VAR-XGBoost model based on Vector autoregression (VAR), Kriging method and XGBoost (Extreme Gradient Boosting). China is used as an example and its spatial distribution of O_3 is simulated. In this paper, the O_3 concentration data of the monitoring sites in China are obtained, and then a spatial prediction method of O_3 mass concentration based on the VAR-XGBoost model is established, and finnally its influencing factors are analyzed. This paper concludes that O_3 features the highest correlation with $PM_{2.5}$ and the lowest correlation with SO_2. Among the measurement factors, wind speed and temperature are the most important factors affecting O_3 pollution, which are positively correlated to O_3 pollution. In addition, precipitation is negatively correlated with 8-hour ozone concentration. In this paper, the performance of the VAR-XGBoost model is evaluated based on the ten-fold cross-validation method of sample, site and time, and a comparison with the results of XGBoost, CatBoost (categorical boosting), ExtraTrees, GBDT (gradient boosting decision tree), AdaBoost (adaptive boosting), RF (random forest), Decision tree, and LightGBM (light gradient boosting machine) models is conducted. The result shows that the prediction accuracy of the VAR-XGBoost model is better than other models. The seasonal and annual average R^2 reaches 0.94 (spring), 0.93 (summer), 0.92 (autumn), 0.93 (winter), and 0.95 (average from 2016 to 2021). The data show that the applicability of the VAR-XGBoost model in simulating the spatial distribution of O_3 concentrations in China performs well. The spatial distribution of O_3 concentrations in the Chinese region shows an obvious feature of high in the east and low in the west, and the spatial distribution is strongly influenced by topographical factors. The mean concentration is clearly low in winter and high in summer within a season. The results of this study can provide a scientific basis for the prevention and control of regional O_3 pollution in China, and can also provide new ideas for the acquisition of data on the spatial distribution of O_3 concentrations within cities.

1. Introduction

Ozone is an ambient air pollutant produced by a non-linear, complex photochemical reaction between atmospheric nitrogen dioxide (NOx) and volatile organic compounds (VOCs) (USEPA, 2013). Ozone is involved in other chemical reactions in the atmosphere and is characterized by short survival period and high activity (Qin et al., 2017; Liu et al., 2022). With rapid population growth and economic development,

Abbreviations: O_3, Ozone; NOx, Nitrogen dioxide; VOCs, Volatile organic compounds; $PM_{2.5}$, Fine particulate matter; PM_{10}, Inhalable particles; SO_2, Sulfur dioxide; MOE, Ministry of Ecology and Environment; VAR, Vector Autoregression; XGBoost, Extreme Gradient Boosting; CatBoost, Categorical Boosting; GBDT, Gradient Boosting Decision Tree; AdaBoost, Adaptive Boosting; RF, Random Forest; LightGBM, Light Gradient Boosting Machine; LSTM, Long Short-Term Memory; MLP, Multilayer Perceptrons; MLR, Multiple Linear Regression; DNN, Deep Neural Networks; SVM, Support Vector Machines; UV, Ultraviolet; PD, Population Density; ED, Economic Density; BTH, Beijing, Tianjin and Hebei; YRD, Yangtze River Delta; PRD, Pearl river delta; CNEMC, China National Environmental Monitoring Center; IRF, Impulse response function; CNN, Convolutional Neural Network; LUR, Land use regression; WRF, Weather Research and Forecasting; ML, Machine learning; ADF, Augmented Dickey-Fullertest; SVR, Support Vector Machine Regression; RMSE, Root mean squard error; MAE, Mean absolute error; MAPE, Mean absolute percentage error; R^2, Coefficient of determination.

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https://doi.org/10.1016/j.ecoenv.2023.114960
Received 27 August 2022; Received in revised form 16 April 2023; Accepted 24 April 2023
Available online 26 April 2023

ozone pollution has become a major atmospheric problem in many Chinese cities, affecting air quality and human health, especially megacities where human activities are high. According to the monitoring results released by the Ministry of Ecology and Environment of China, from 2013 to 2017, the average annual average daily maximum 8-hour ozone concentration in 74 cities in China is 139–167 mg/m^3 ((Ministry of Ecology and environment of China, 2022). According to monitoring data, ozone concentration in China has been gradually increasing in recent years. Beijing, Tianjin and Hebei (BTH), the Yangtze River Delta (YRD) and the Pearl River Delta (PRD) are experiencing severe ozone pollution events (Chen et al., 2022a). O_3 plays a protective role for humans and the environment in the stratosphere, but if O_3 continues to increase in the troposphere, it will have a great impact on human health and the ecosystem, such as eyes, respiratory tract and lungs (Fann et al., 2012; Wang et al., 2020). In China, O_3 is estimated to increase in the ground-level ozone layer, causing 120,000 premature deaths, 36,000 cardiovascular deaths and 18,000 respiratory deaths each year (Liang et al., 2019). High concentration of ozone pollution affect health, including morbidity and mortality caused by respiratory and cardiovascular diseases (Bello-Medina et al., 2019). What's more, it also affects plant growth and crop production (Yi et al., 2020).

Therefore, accurate prediction of O_3 concentration is of practical importance for air pollution control and sustainable economic development. An increasing number of machine learning and deep learning methods are being applied to atmospheric pollutant concentration prediction, such as LSTM (long and short-term memory networks) (Shi et al., 2022), MLP (multilayer perceptrons) (Dai et al., 2022a), MLR (multiple linear regression) (Borlaza et al., 2021), LightGBM (Zhong et al., 2021), XGBoost (Wang et al., 2022a). These models offer better performance in predicting atmospheric pollutants than traditional statistical models, because it can capture the non-linear characteristics of the data (Dai et al., 2022b). Random forests are widely used in O_3 concentration prediction. Zhan et al. (2018) developed a random forest model to predict the maximum daily 8-hour average ozone concentration for human exposure assessment in China. In 2015, it used an air quality monitoring network, which has been developed by using meteorological, elevation and emission inventory data in recent years. Zhu et al. (2022) used a random forest framework to predict monthly mean daily maximum 8-hour average (MDA8) O_3 concentrations in China from 2005 to 2019. Ma et al. (2021) developed a high-performance random forest model with a resolution of $0.01^\circ\times\ 0.01^\circ$ based on training data from 2013 to 2017 in the Beijing-Tianjin-Hebei (BTH) region of China and he estimated the daily maximum 8-hour mean O_3 (O_3-8hmax) concentration. More various supervised machine learning algorithms are gradually being used for ozone concentration prediction, such as the linear algorithm SVR (Su et al., 2020), Land use regression (LUR) (Ren et al., 2020) and the nonlinear algorithms SVM (Shi et al., 2021), DNN (Chen et al., 2022b). Yafouz et al. (2021) used independent machine learning models (MLP, SVR, and XGBoost) and deep learning methods (LSTM and CNN) to predict the reliability of O_3 concentration. Both methods are characterized by effectively predicting hourly ozone trends with fewer input variables. At present, the research on the spatiotemporal characteristics of ozone focuses on a small number of cities, such as systematic study of the temporal and spatial changes of air pollutants in Beijing, Tianjin and Hebei (BTH) area from 2014 to 2021. Lyu et al. (2022). Li et al. (2022) analyzed the spatial and temporal law of O_3 concentrations in Beijing based on the simulation results of meteorological research and forecasting (WRF) and environmental monitoring data in Beijing and they found that the XGBoost algorithm was able to better capture the spatial and temporal patterns of pollutant concentrations. However, too much manual involvement is not only time-consuming and labor-intensive, but also too complicated for the engineering implementation of the model. In addition, the current model, despite of high efficiency in reducing the complexity of the objective function, ignores the spatial and temporal variability of O_3 concentrations. At present, domestic and international scholars have mainly analyzed the spatial and temporal distribution characteristics of O_3 in specific urban clusters (Wang et al., 2021), single cities (Chen et al., 2019), or single-year changes (Zhang et al., 2020), with relatively few analyses involving national-scale and multi-year changes. The number of ground-based monitoring stations is usually limited and they can only reflect local pollutant concentrations, which cannot reveal the spatial heterogeneity of O_3 concentrations within a large study area. This poses a great challenge to the spatial characterization of O_3 pollution. The use of related models is still relatively uncommon and relies heavily on manual operation. Existing machine learning models has limitations in considering the spatio-temporal characteristics of atmospheric O_3 itself. Currently, existing machine learning models haven't performed well in considering the spatiotemporal characteristics of atmospheric O_3. For instance, the existing tree models use their own decision tree algorithms to filter features, but ignore the relationships among individual influencing factors. What's more, they are unable to estimate the dynamic relationships of all endogenous variables. Therefore, it is urgent to promote machine learning models that take into account the analysis of influencing factors and spatial-temporal heterogeneity.

The contributions of this study are as follows:

(1) The inversion of O_3 helps to capture the regional variability of O_3 processes in time and space. Therefore, with the help of the inversion of O_3, it is helpful to grasp the regional change process of O_3 in time and space. Specifically, by gradually selecting the VAR model to determine the influencing factors, a predictive model using the eight-tree model is developed to improve the accuracy of O_3 change prediction.
(2) The hybrid spatial forecasting model proposed in this paper combines the advantages of a VAR model to estimate the dynamic relationships of all endogenous variables. VAR models can predict interconnected time-series systems and can analyse the dynamics of shocks to a system of variables from random perturbations, as a result, it explains the impact of individual shocks. It is a good remedy for the shortcomings of some existing atmospheric system models, not only for modelling the interrelationships and interactions between the structure and function of atmospheric system, but also for the advantage of simple variable determination. Hybrid spatial prediction models that identify the most influential emission predictors and the predictability of tree models in estimating non-linear trends would be more widely effective than traditional tree model estimation methods. Validated by R^2, RMSE and MAE metrics, the results show that VAR-XGBoost performs better.
(3) This paper proposes a VAR-tree model to estimate the spatial and temporal distribution of O_3 concentrations based on Chinese O_3 observation data from 2016 to 2021, data on atmospheric pollutants, conventional meteorological observation elements, and anthropogenic data. In this paper, a concentration model of the four seasons and the average concentration model from 2016 to 2021 are established in China, and the spatial and temporal characteristics of O_3 concentration in China are analyzed. By analyzing the spatial and temporal distribution patterns of O_3 and its influencing factors, the changes in O_3 at different time scales and their underlying mechanisms in recent years are elucidated, and the general pattern of spatial and temporal distribution of O_3 concentrations in China is summarized.

2. Data sources and methods

2.1. Data description

2.1.1. Study area

Fig. 1 shows the distribution of all monitoring stations in China. To analyze the spatial and temporal variation of ozone obviously, 361 cities

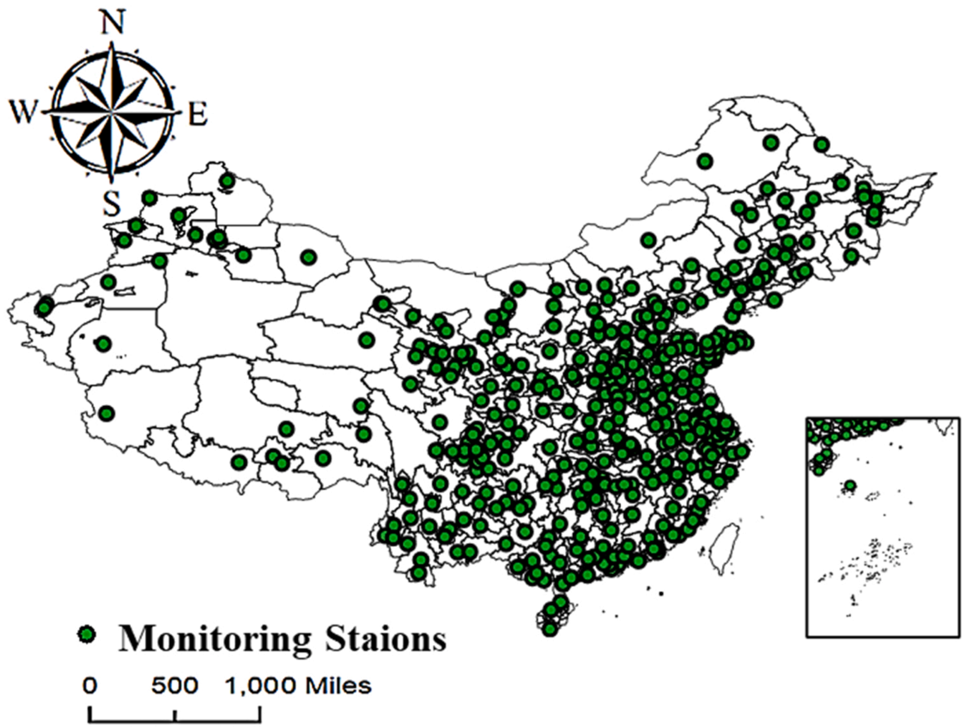

Fig. 1. Distribution of ground monitoring stations in China.

in China are considered in the 2016–2021 China O_3 Spatial and Temporal Distribution Survey.

2.1.2. Air pollutant data

The monitoring data of $PM_{2.5}$, PM_{10}, SO_2, CO and NO_2 from 1st January 2016–31 st December 2021 used in this study are obtained from the automatic monitoring network of China National Environmental Monitoring Center (CNEMC). All pollutant data are monitored in accordance with the automatic monitoring standards for the air environment.

2.1.3. Meteorological data

The meteorological factors selected for this paper are shown in Table 1. To further investigate the influence of meteorological conditions on ozone production, this paper applies data from monitoring stations at the National Meteorological Science Data Centre (http://data.cma.cn/) in China. The main meteorological factors includes relative humidity, air temperature, surface pressure, wind speed, precipitation, barometric pressure, lighting market.

2.1.4. O_3 data

The O_3 data are derived from the OMI sensor of Earth observation satellite Aura (a polar orbiting satellite in sun-synchronous orbit launched on 15 July 2004). The daily ozone column concentration data selected for 2016–2021 in this paper are derived from the total number of ozone columns on the NASA website. The data are a level 3 0.25° resolution from the OMI OMTO3e grid product in the NASA website (source of data information: https://disc.gsfc.nasa.gov/).

2.1.5. Human activity data

Data of economic density (ED) and population density (PD) are from the Data Center for Resource and Environmental Sciences, Chinese Academy of Sciences, 2016–2021 (https://www.resdc.cn/Default.aspx).

2.2. Data division

According to 4:1 ratio, first of all, we divide the data set into two parts, and one is a training set and another is a test set. The training set data are used to build the model. The model is then evaluated by using a 10-fold cross-validation approach, where the sample data are randomly divided into 10 copies, with 9 copies randomly selected as the training set and another one as the test set. When the round is completed, 9 copies are randomly selected again to test the data. After several rounds (less than 10), a loss function is selected to evaluate the optimal model and parameters. We use data from the test set for external validation. Testing the accuracy of an already trained model confirms the actual

Table 1
Meteorological factors and abbreviations.

Variable	Unit	Abbreviation	Variable	Unit	Abbreviation
Average daily surface temperature	℃	AvgST	Average temperature	℃	AvgT
Daily maximum surface temperature	℃	HighST	Highest Temperature	℃	HighT
Daily minimum surface temperature	℃	LowST	Lowest Temperature	℃	LowT
Average wind speed	m/s	Avg(m/s)	Hours of sunshine	h	Sunshine(h)
Maximum wind speed	m/s	High(m/s)	Average Humidity	%	Avg(%)
Highest wind direction	-	Highdirection	Lowest Humidity	%	Low(%)
Extreme wind speeds	m/s	Extrem(m/s)	Average air pressure	hPa	Avg(hpa)
Precipitation from 20 to 8 pm	mm	20–8(mm)	Highest air pressure	hPa	High(hpa)
Precipitation from 08 to 20 h	mm	8–20(mm)	Lowest air pressure	hPa	Low(hpa)
Accumulated precipitation at 20–20 h	mm	20–20(mm)	Season	-	Season

predictive power of the network.

2.3. VAR model

VAR model was proposed by Sims in 1980. It differs from traditional econometric methods because it is not based on economic theory. However, on the form of a multi-party linkage (Sims, 1980), VAR models are commonly used to forecast interconnected time series systems and to analyse the dynamic shocks of stochastic perturbations to a system of variables, so as to explain the effects of various shocks on the formation of variables.

The VAR model equation is as follows :

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t t = 1, 2, \ldots, T \tag{1}$$

2.3.1. ADF test

Traditional VAR models obey the time series to be smooth or co-integrated to avoid the "pseudo-regression" problem. In such cases, for instance, there is no systematic change in the mean and variance of the time series, and there is no strictly eliminated periodic change, the time series is a stationary time series.

The main methods are the graphical test method and the square root method, and this paper mainly applies the square root method. The square root test is used to test the smoothness of an autoregressive equation by checking whether the characteristic roots of the equation are all within the unit root. The most commonly used test is the Augmented Dickey-Fullertest (ADF) test, where the null hypothesis (H0) is the series that has a unit root and non-stationary. The autoregressive equation AR(p) for any p-order is shown below:

$$Y_t = \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2} + \cdots + \varphi_p Y_{t-p} + \varepsilon_t, t = 1, 2, \ldots, T \tag{2}$$

Y_t is the time series of air pollution and meteorological indicators to be tested in this paper. t is the time delay of variable Y. p is the order of the autoregressive model. $\varphi_i (i = 1, 2, 3, \ldots, p)$ is the autoregressive model coefficient. ε_t is the white noise series with mean 0. Eq. (2) is a p-order autoregressive process with no constant mean and no trend. It can be expressed as Eq. (3):

$$\lambda^p - \varphi_1 \lambda^{p-1} - \varphi_p = 0 \tag{3}$$

$\lambda^i (i = 1, 2, \ldots, p)$ is the feature root. The series is smooth if all features heels of the equation lie within the unit circle, i.e. $|\lambda^i| < 1$.

2.3.2. Generalized impulse response function

Because the VAR model is a non-theoretical model, when we analyze the VAR model, we cannot analyze the impact of the change of one variable on another variable, instead, we need to analyze the impact of a change in an error term, or the dynamic impact on the system when the model is subjected to some kind of impact. Therefore, the impulse response function (IRF)2 needs to describe the effect of an information shock of its own standard deviation of random error on the current and future values of other variables, and it can also characterize the direction and duration of this effect. However, due to the traditional orthogonal impulse response function, different results are produced depending on the order of the variables. Based on Eq. (3), one of the variables of the shock is selected and the generalised impulse response function for the impact of the other shocks t. is calculated. The generalized impulse response function at this moment. It can be expressed as Eq. (4):

$$GI_y\left(n, \varphi, \Omega_{t-1}\right) = \sqrt{\sigma_{ii}} A_n \sum e_i \tag{4}$$

φ is the k-dimensional shock quantity. $n = 1, 2, \cdots N$ is the number of periods of the impulse response. Ω_{t-1} is the set of relevant information over period $t-1$. $GI_y(n, \varphi, \Omega_{t-1})$ is the effect of a φ shock on the expected value of Y_{t+n} resulting from an impulse response function value $\sqrt{\sigma_{ii}} = \varphi_i$. $\varphi = E(\xi_t | \xi_{it}) = \sqrt{\sigma_{ii}}$; e_i is a $k \times 1$-dimensional selection matrix, with the ith element being 1 and the other elements being 0.

2.4. The Kriging principle

The basic principle of Kriging's method is to estimate data at other unobserved locations in space from data at regularly distributed sample points in space (Sampson et al., 2013).

- Regional variables: the study area can be considered as a regional variable $Q(R)$. R_1, R_2,., R_n are the O_3 observations at the corresponding stations in the area. $Q(R_1)$, $Q(R_2)$,., $Q(R_3)$ are the O_3 observation value of the corresponding station. For R_0 in the region, the spatial value $Q_d(R_0)$ can be obtained by Kriging's method of interpolation and the temporal value Q_t can be expressed in terms of the month in which the point is located. It can be expressed as Eq. (5) (6):

$$Q_d(R_0) = \sum_{i=1}^{n} \omega_i Q(R_i) \tag{5}$$

$$Q_t = m \tag{6}$$

$Q_d(R_0)$ is the spatial value of a given point. ω_i is the Kriging weight. $Q(R_i)$ is the station monitoring value around the point, and m is the month of a given point.

Kriging meets a set of optimal coefficients with the smallest difference between the predicted value and the true value at the monitoring site, and it also meets the condition of non-deviation from the estimated data. It can be expressed as Eq. (7) (8):

$$\min_{\omega i} Var\left(Q_d(R_0) - Q(R_0)\right) \tag{7}$$

$$E(Q_d(R_0) - Q(R_0)) = 0 \tag{8}$$

- Variogram: variogram is the basis of the kriging interpolation method. It is a model function used to describe the spatial relationship among O_3 ground monitoring stations or relationship between monitoring stations and pixels. Variogram of the regional variable $Q(R)$ can be expressed as the semi-variance $\varepsilon(R_i, R_j)$ of the difference between the observations at sites R_i and R_j.It can be expressed as Eq. (9):

$$\varepsilon\left(R_i, R_j\right) = \frac{1}{2} E\left[Q_d(R_i) - Q(R_j)\right]^2 \tag{9}$$

- Solution to the equations: The Kriging equations can be obtained as Eqs. (10) and (11) in the case of unbiased and minimal estimated variance.

$$\sum_{i=1}^{n} \omega_i \varepsilon(x_i, x_j) - \varphi = \varepsilon\left(x_0, x_i\right) \tag{10}$$

$$\sum_{i=1}^{n} \omega_i = 1 \tag{11}$$

In this equation, φ is the Lagrangian multiplier factor. By solving the above equation, we can get the Kriging weights ω_i, which in turn yields an estimate of any point R_0 in the region $Q_d(R_0)$. The Kriging method takes full account of the correlation of the O_3 site data by calculating the variance function of the sample.

2.5. Tree model

2.5.1. Decision trees

Each internal node in a decision tree is a splitting problem: a test for an attribute of the instance is specified, and it splits the samples arriving at that node according to a particular attribute. Each subsequent branch of that node corresponds to one of the possible values of that attribute (Burduk, Biedrzycki, 2022). The mean of the output variables in the samples contained in the leaf nodes of the regression tree is the predicted outcome.

2.5.2. Random forest

Random forest regression refers to the random sampling of sample observations and feature variables of the modeled data set in the process of generating many decision trees (Utkin, Konstantinov, 2022). The test results for each sample are one tree. Each tree generates rules and judgment values that match its own attributes. Finally, the forest integrates the rules and judgment values of all decision trees in order to achieve the regression of the random forest algorithm.

2.5.3. AdaBoost

Adaboost gives a high weight to the learner with a low error rate, and gives a low weight to the learner with a high error rate, and then combines the weak learner and the corresponding weight, and finally generates a strong learner (Liu et al., 2022). The difference between the regression problem algorithm and the classification problem algorithm is the way the error rate is calculated. Classification problem algorithms generally use 0/1 loss function, while regression problem algorithms generally use square loss function or linear loss function.

2.5.4. GBDT

The GBDT model is an additive model that trains a set of CART regression trees serially and eventually sums the predictions of all the regression trees, resulting in a strong learner where each new tree fits the negative gradient direction of the current loss function (Zhang et al., 2022). Finally, the summation of this set of regression trees is output to obtain the regression results.

The model is a one-time iteration variable. During iteration, it increases the submodels one by one, and ensures the loss function is constantly decreasing. That is, the residuals are approximated by the negative gradient of the loss function at the value of the current model $f(x) = f_{j-1}(x)$. The number of training samples is i. The number of iterations is j. The loss function is $L(y_i, f(x_i))$. Then the negative gradient r_{ij} is calculated:

$$r_{ij} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f(x)=f_{j-1}(x)} \tag{12}$$

A regression tree is drafted according to r_{ij} to obtain the number of leaf nodes j in the region R_{jk} and$k = 1, 2, \ldots, k$. k is the number of leaf nodes.

For $k = 1,2,\ldots,k$, it uses linear search to estimate the value of the leaf node region C_{jk}, so that the loss function is minimized as follows:

$$C_{jk}=\underset{C}{\operatorname{argmin}}\sum_{x_i\in R_{jk}} L\left[y_i, f_{j-1}(x_i)+C\right] \tag{13}$$

Update the regression tree: I is the indicator function that takes the value of 1 when the regression tree determines that $x \in R_{mj}$ and 0 otherwise.

$$f_j(x) = f_{j-1}(x) + \sum_{k=1}^{k} C_{jk} I\left(x \in R_{jk}\right) \tag{14}$$

Output the final model after j times:

$$\hat{f}(x) = f_j(x) = \sum_{j=1}^{J}\sum_{k=1}^{k} C_{jk} I\left(x \in R_{jk}\right) \tag{15}$$

2.5.5. Extra-trees

Extra-Trees (extremely random forests) are very similar to random forests, but here the "randomness" is expressed in the node partitioning of the decision trees (Hui et al., 2023), which simply uses random features and random thresholds, so that each of our decision trees has a larger and more random shape and variation.

2.5.6. CatBoost

CatBoost is a GBDT framework based on the symmetric decision tree algorithm (Zeng et al., 2023). It performs well in three parts, efficient and reasonable processing of category-based features, handling of gradient bias and prediction bias problems to improve the accuracy and generalization of the algorithm.

2.5.7. XGBoost

XGBoost is an efficient implementation of GBDT (Ye et al., 2023). Unlike GBDT, XGBoost adds a regularization term to the loss function; Because some loss functions are difficult to calculate the derivative, XGBoost uses the second-order Taylor of the loss function to fit the loss function. The detailed parameters are shown in Table S1 as follows:

2.5.8. LightGBM

The idea behind LightGBM is to discrete continuous floating-point features into k values and construct a Histogram of width k. (Wang et al., 2022b) The training data are then traversed and the cumulative statistics for each discrete value in the histogram are calculated. In feature selection, only the discrete values of the histogram need to be traversed to find the optimal segmentation points; and the use of a leaf-wise strategy with depth restrictions saves a lot of time and space.

2.6. Research framework

Fig. 2 shows a diagram of the structure of the model for this study. The fixed parameters selected for this study are the Chinese region. Variable parameters include air quality data and meteorological data obtained from China's National Environmental Monitoring Centre. Two methods of variable selection are used in this study. After discussing the ozone impact factors with the help of the VAR model and selecting statistically significant predictor variables that meet the criteria, these variables are then applied to the eight tree model algorithm. A total of eight models are developed in this study (i.e. VAR-DecisionTree, VAR-ExtraTrees, VAR-RF, VAR-CatBoost, VAR-AdaBoost, VAR-LightGBM, VAR-GBDT and VAR-XGBoost). Secondly, data partitioning, 10-fold cross-validation, external data validation and seasonal and year-based validation methods are used to validate the robustness of the developed models. Finally, the spatial and temporal distribution patterns of surface ozone in China are analyzed by applying GIS and improved Kriging methods.

3. Results

3.1. VAR model results

3.1.1. VAR model steps

(1) The VAR model needs to be tested for the smoothness of each time series variable before it is built. If the time series are smooth, then a VAR model can be built, otherwise the resulting vector autoregressive model is a pseudo-regression. If $p < 0.05$, then the series is a smooth series. If $p >= 0.05$, then the series is non-stationary. A vector autoregressive model can also be built if the individual data do not meet the stationarity requirement, as long as it pass the co-integration test.
(2) Comparison of different lag orders (According to the information of different hysteresis orders, a better hysteresis order is found, and then the VAR model is re-established.). As shown in Fig. 3 , it can be seen that a positive shock of a standard deviation size to O_3 brings a negative increase in $PM_{2.5}$ concentration in the first

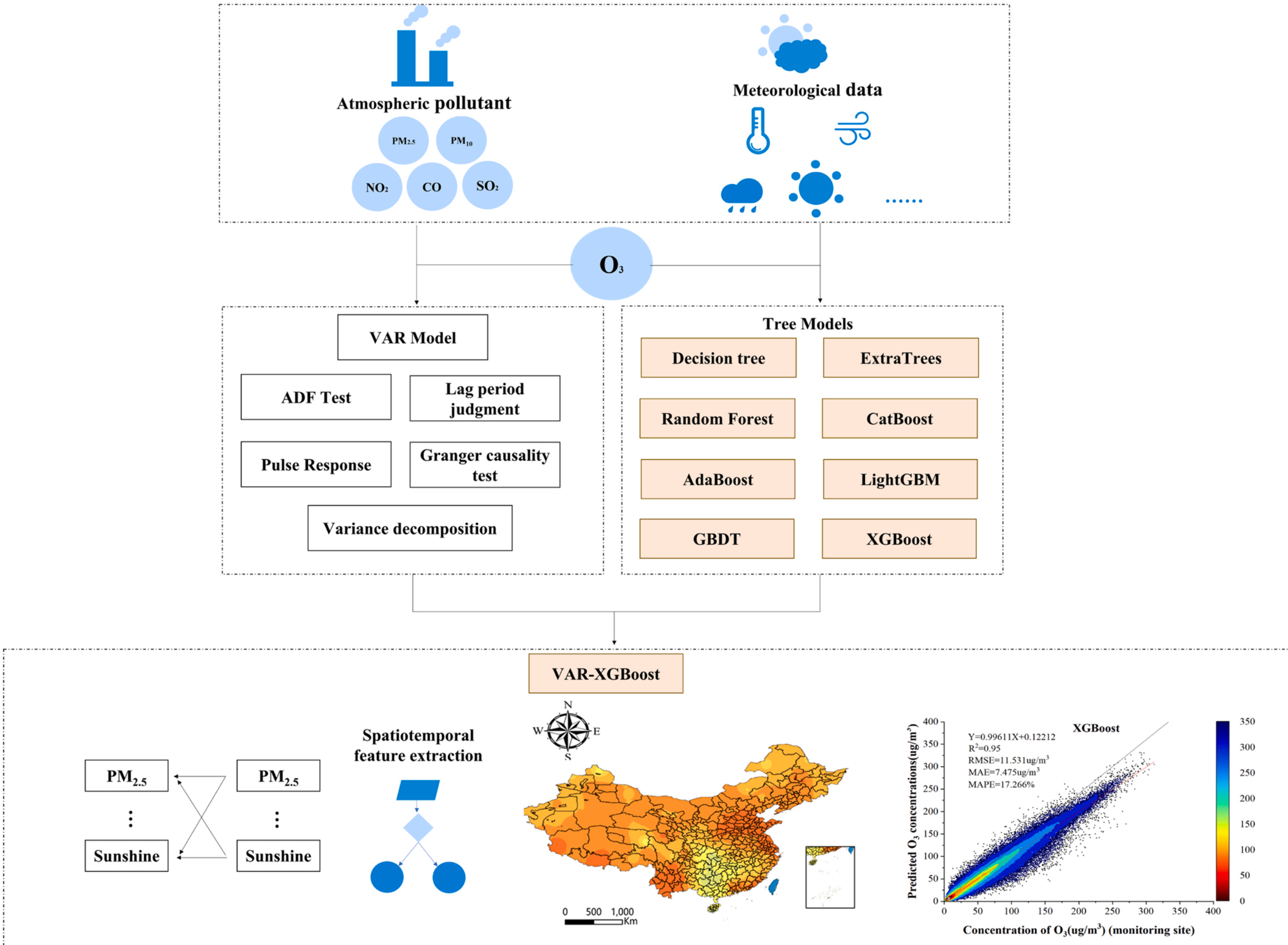

Fig. 2. The model structure of the spatio-temporal characteristics of O3 concentrations in China based on VAR-tree model.

period and a fluctuating increase in $PM_{2.5}$ concentration with a maximum after the second lag period.

(3) Build a VAR model and estimate the parameters.

(4) After the VAR model is built, the model needs to be tested for stability. Only after the test is passed can impulse response analysis and variance decomposition be performed.

3.1.2. *VAR vector autoregressive model results*

Table 2 shows the results of the ADF test, including variables, t-test results, and AIC values, for testing whether the time series is smooth.

Table S2 shows the information criterion for the vector autoregressive model with lags of order *p* for selecting the better lag order. It includes logL, FPE, AIC, SC and HQ, where logL is involved in the calculation of FPE, AIC, SC and HQ, which are finally evaluated by the indicators of FPE, AIC, SC and HQ. There are two rules for choosing the optimal lag order: rule No. 1: If there is the most * for a particular lag order, it is recommended to choose that lag order to build the VAR model; Rule No. 2: If there are orders with the same number of * , then it is recommended to choose the smallest possible order.

From the results of the four evaluation indicators of FPE, AIC, SC and HQ, the lag order is recommended to be chosen as the 2nd order and a VAR(2) model is established.

O_3 = 1.683 *O_3(−1)− 0.796 *O_3(−2)+ 0.288 *SO_2(−1)+ 0.014 *SO_2(−2)+ 0.433 *avg(℃)(−1)− 0.214 *avg(℃)(−2)+ 0.041 *high (℃)(−1)− 0.049 *high(℃)(−2)− 0.344 *low(℃)(−1)+ 0.401 *low (℃)(−2)+ 1.381 *avg(m/s)(−1)− 0.649 *avg(m/s)(−2)+ 0.124 *high (m/s)(−1)− 0.528 *high(m/s)(−2)− 0.056 *highdirection(−1)− 0.024 *highdirection(−2)− 0.017 *extrem(m/s)(−1)+ 0.039 *extrem (m/s)(−2)+ 0.102 *extremdirection(−1)− 0.084 *extremdirection (−2)− 5.763 * 20–8(mm)(−1)+ 8.557 * 20–8(mm)(−2)− 5.799 * 8–20(mm)(−1)+ 8.527 * 8–20(mm)(−2)+ 0.395 *CO(−1)+ 2.148 *CO(−2)+ 0.293 *NO_2(−1)− 0.361 *NO_2(−2)+ 5.777 * 20–20 (mm)(−1)− 8.535 * 20–20(mm)(−2)− 0.02 *$PM_{2.5}$(−1)+ 0.013 *$PM_{2.5}$(−2)− 1.461 *avg(℃).1(−1)+ 1.379 *avg(℃).1(−2)+ 0.023 *PM_{10}(−1)− 0.018 *PM_{10}(−2)+ 0.132 *high(℃).1(−1)− 0.123 *high(℃).1(−2)+ 0.879 *low(℃).1(−1)− 0.931 *low(℃).1 (−2)− 0.085 *sunshine(h)(−1)+ 0.214 *sunshine(h)(−2)− 0.098 *avg (%)(−1)+ 0.088 *avg(%)(−2)+ 0.022 *low(%)(−1)− 0.035 *low(%) (−2)− 0.943 *avg(hPa)(−1)+ 1.256 *avg(hPa)(−2)+ 0.662 *high (hPa)(−1)− 0.874 *high(hPa)(−2)+ 0.272 *low(hPa)(−1)− 0.426 *low(hPa)(−2)+ 52.503.

3.1.3. *Impulse response results*

Fig. 3 shows the impulse response analysis. It describes the impact of a shock to one endogenous variable (the shock variable) on another endogenous variable (the shocked variable) in a VAR model. It can be observed that when we give SO_2, NO_2, PM_{10} and CO a positive impact of standard deviation size, the O_3 concentration will increase in the short term. But this effect will gradually decrease over time. In the impulse response from O3 to SO2 and NO2, the $PM_{2.5}$ concentration reaches a maximum with 4–6 hysteresis. The impulse response for O_3 and $PM_{2.5}$ shows that the increase in $PM_{2.5}$ concentration stabilizes around the 9th lag period for $PM_{2.5}$ concentration, and it can be assumed that the effects of $PM_{2.5}$, SO_2, NO_2, PM_{10} and CO concentrations are long-lasting and

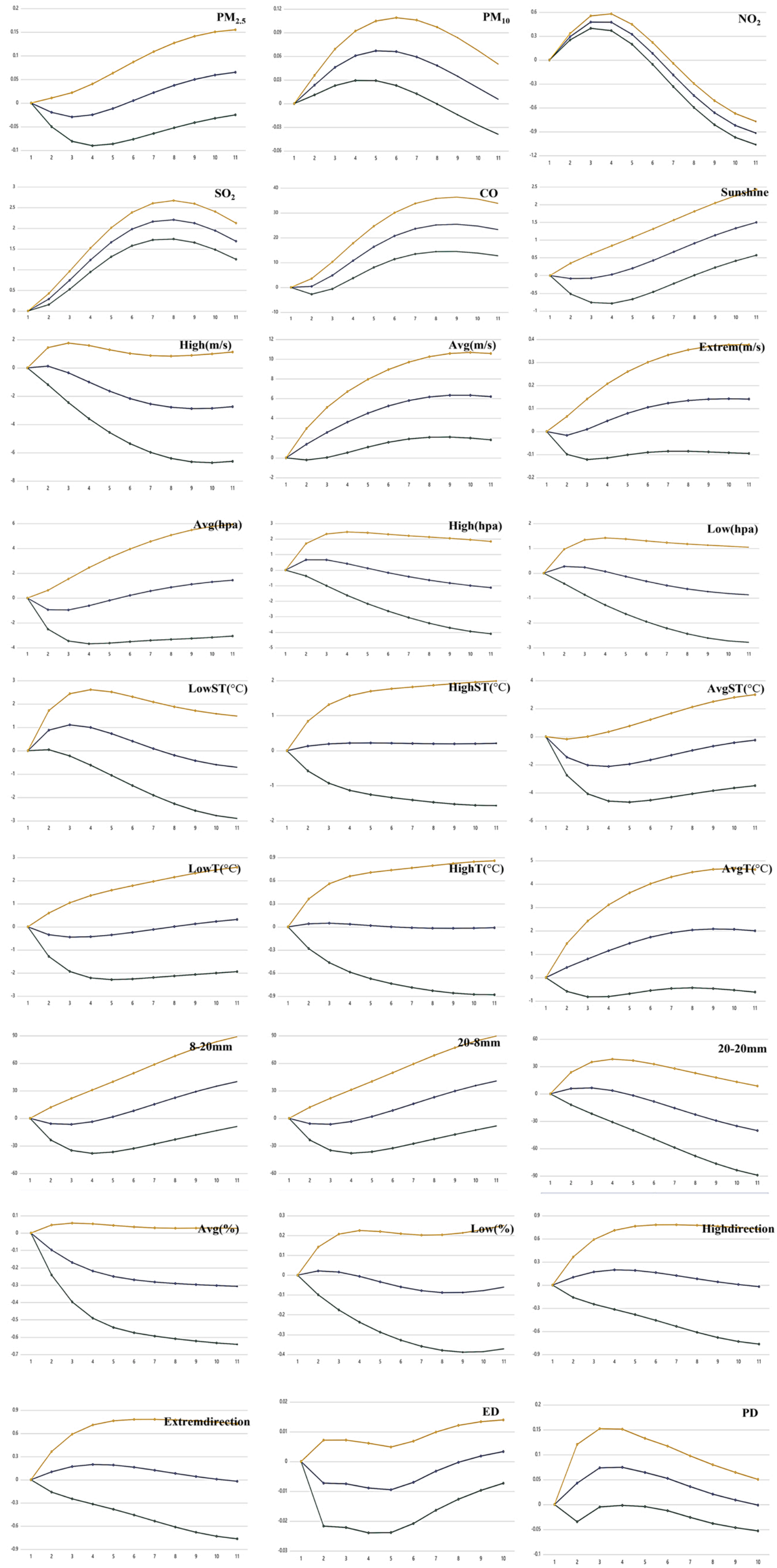

Fig. 3. Impulse response of O_3 to atmospheric pollutants and meteorological factors.

Table 2
Augmented Dickey-Fuller Unit Root Test for O_3 and its Influencing Factors.

Variables	t-statistic	p-value	Threshold values		
			1%	5%	10%
O_3	-5.859	< 0.001 * **	-3.431	-2.862	-2.567
High(hpa)	-3.961	0.002 * **	-3.431	-2.862	-2.567
20–8(mm)	-10.318	< 0.001 * **	-3.431	-2.862	-2.567
20–20(mm)	-9.987	< 0.001 * **	-3.431	-2.862	-2.567
Extrem(m/s)	-8.43	< 0.001 * **	-3.431	-2.862	-2.567
Low(hpa)	-4.682	< 0.001 * **	-3.431	-2.862	-2.567
$PM_{2.5}$	-8.571	< 0.001 * **	-3.431	-2.862	-2.567
Highdirection	-10.298	< 0.001 * **	-3.431	-2.862	-2.567
Sunshine(h)	-8.36	< 0.001 * **	-3.431	-2.862	-2.567
Avg(m/s)	-8.618	< 0.001 * **	-3.431	-2.862	-2.567
Extremdirection	-10.262	< 0.001 * **	-3.431	-2.862	-2.567
Low(%)	-6.072	< 0.001 * **	-3.431	-2.862	-2.567
High(T)	-2.338	0.160	-3.431	-2.862	-2.567
Avg(T)	-1.803	0.379	-3.431	-2.862	-2.567
8–20(mm)	-10.494	< 0.001 * **	-3.431	-2.862	-2.567
Avg(ST)	-12.299	< 0.001 * **	-3.431	-2.862	-2.567
Low(T)	-2.189	0.210	-3.431	-2.862	-2.567
Avg(%)	-5.53	< 0.001 * **	-3.431	-2.862	-2.567
Avg(hpa)	-4.688	< 0.001 * **	-3.431	-2.862	-2.567
Low(ST)	-2.319	0.166	-3.431	-2.862	-2.567
High(ST)	-3.344	0.013 * *	-3.431	-2.862	-2.567
NO_2	-8.804	< 0.001 * **	-3.431	-2.862	-2.567
Economic density	-3.622	0.005 * **	-3.589	-2.93	-2.603
Population density	-1.542	0.513	-3.571	-2.923	-2.599

Note: * ** , * * represent 1%, 5% level of significance respectively

positively related, reflecting that the air quality system is closely linked. At the same time, the control of $PM_{2.5}$ requires the cooperation of various pollution control measures for SO_2, NO_2, PM_{10} and CO to have good results. It can be seen that a positive shock of a standard deviation size to O_3 brings a negative increase in $PM_{2.5}$ concentration in the first period and a fluctuating increase in $PM_{2.5}$ concentration with a maximum after the second lag period. However, the $PM_{2.5}$ concentration shifts to a positive increase after the 4th lag period, indicating that the overall effect of O_3 on $PM_{2.5}$ concentration is negative and O_3 has a suppressive effect on $PM_{2.5}$ concentration. It also can be seen that the effect of O_3 on economic density increases in the first period, decreases and reaches a minimum after the second cycle, and then it gradually increases and slows down after the fifth cycle. The effect of O_3 on population density starts to increase in the first cycle and reaches a maximum in the third cycle and then gradually decreases. It can be assumed that the overall effect of O_3 on economic density is positive, while the effect on population density is negative.

It can be seen from Fig. S1 that the inverse roots of the AR characteristic polynomials are all within the unit circle. And the air quality system model consisting of $PM_{2.5}$, PM_{10}, SO_2, CO, NO_2 and O_3 are stable in the long term, indicating that the VAR(2) model can be used for subsequent studies.

3.1.4. Granger causality test

Granger causality tests are mainly used to analyse the causal relationship between variables, and to determine whether a change in one variable is the cause of a change in another one. The Granger causality test determines the extent to which the variable y can be explained by past values of the variable x, that is, whether inputing lag period of the variable x improves the explanatory power or not. Granger causality between x and y holds if the variable y is affected by a lag period of the variable x. The original hypothesis of the Granger causality test is that variable x is not the Granger cause of variable y. The alternative hypothesis is that variable x is the Granger cause of variable y. The results of the test are shown in Table S3.

3.1.5. Variance decomposition results

Table 3 shows the variance decomposition results. The variance decomposition is an analysis of the proportion of the standard deviation of the forecast residuals affected by different shocks, and corresponds to the proportion of the endogenous variables contributing to the standard deviation.

To further analyse the impact of PM_{10}, SO_2, NO_2, CO and O_3 on $PM_{2.5}$, the variance decomposition of O_3 based on the VAR (2) model is carried out after selecting 1–10 lags. The results are shown in Table 3. O_3 concentration is only influenced by itself at lag 1. The impact of SO_2 and NO_2 changes on O_3 concentration intensifies during the inspection

Table 3
O_3 Proportion of contribution of endogenous variables to the results of standard deviation decomposition.

Number of steps	1	2	3	4	5	6	7	8	9	10
Standard deviation	10.64	17.92	23.52	27.83	31.09	33.43	34.9	35.88	36.37	36.63
O_3%	100	99.58	98.72	97.661	96.505	95.284	94.15	93.132	92.242	91.389
$PM_{2.5}$%	0	0.065	0.137	0.277	0.464	0.731	1.028	1.325	1.574	1.726
PM_{10}%	0	0.084	0.192	0.314	0.426	0.544	0.626	0.668	0.669	0.66
NO_2%	0	0.231	0.782	1.306	1.711	1.919	1.95	1.883	1.845	2.009
CO%	0	0.005	0.056	0.226	0.55	1.032	1.616	2.236	2.797	3.214
PD%	0	0.392	2.161	2.747	2.968	2.878	2.814	2.781	2.723	2.652
ED%	0	1.515	2.119	2.195	2.788	2.9	2.819	2.748	2.748	0
SO_2%	0	0.011	0.054	0.103	0.159	0.208	0.231	0.247	0.25	0.248
Avg(ST)%	0	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.005	0.006
High(ST)%	0	0.001	0.006	0.008	0.01	0.013	0.017	0.018	0.018	0.029
Low(ST)%	0	0.001	0.004	0.009	0.016	0.024	0.032	0.036	0.038	0.041
Avg(m/s)%	0	0.002	0.009	0.028	0.047	0.072	0.104	0.132	0.168	0.216
High(m/s)%	0	0.001	0.002	0.005	0.008	0.012	0.016	0.02	0.024	0.027
Highdirection%	0	0.001	0.003	0.006	0.008	0.012	0.016	0.021	0.027	0.033
Extrem(m/s)%	0	0.001	0.001	0.004	0.009	0.012	0.015	0.018	0.019	0.02
Extremdirection%	0	0.002	0.002	0.002	0.001	0.001	0.001	0.001	0.002	0.003
20–8(mm)%	0	0	0.002	0.004	0.007	0.01	0.013	0.015	0.017	0.017
8–20(mm)%	0	0	0	0.001	0.002	0.002	0.003	0.004	0.005	0.007
20–20(mm)%	0	0	0.001	0.002	0.002	0.002	0.003	0.006	0.008	0.01
Avg(T)%	0	0	0.001	0.003	0.011	0.024	0.045	0.073	0.099	0.121
High(T)%	0	0	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.002
Low(T)%	0	0	0	0	0.001	0.001	0.003	0.005	0.005	0.005
Sunshine(h)%	0	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.002	0.002
Avg(%)%	0	0.006	0.02	0.035	0.048	0.07	0.091	0.115	0.138	0.163
Low(%)%	0	0	0	0.001	0.003	0.005	0.008	0.011	0.012	0.012
Avg(hpa)%	0	0	0	0.002	0.006	0.012	0.019	0.023	0.029	0.031
High(hpa)%	0	0	0	0	0	0.001	0.002	0.002	0.002	0.003
Low(hpa)%	0	0	0.001	0.003	0.003	0.004	0.005	0.005	0.005	0.004

period and reaches 6.333% and 4.954% respectively at lag 10. In general, changes of SO_2 and NO_2 have a greater impact on O_3, which indicates an effective management of SO_2 and NO_2 will help solve the O_3 problem. To further analyze the impact of the five air pollutants and meteorological factors on O_3, the variance decomposition of O_3 based on the VAR (2) model is carried out after selecting 1–10 lags. O_3 concentration is only influenced by itself at lag 1. The impact of CO and NO_2 changes on O_3 concentration intensifies during the inspection period and reaches 3.214% and 2.009% respectively at lag 10. Overall, CO and NO_2 changes have a greater impact on O_3, which indicates an effective management of CO and NO_2 contributes to solving the O_3 problem. Among the meteorological factors, the impact of average wind speed, average temperature and average precipitation over O_3 concentration intensifies. Among the anthropogenic factors, population density has an increasing influence on the O_3 effect. But the value reaches a maximum of 1.968% at lag 5, then decreases slowly. Economic density has an increasing influence on O_3 effect, but the value reaches a maximum of 1.9% at lag 6, then slowly decreases. Overall, economic and population density have obvious effects on O_3.

Table S4 shows the prediction results of the different tree models, showing the four metrics of R^2, RMSE, MAE and MAPE for the eight tree models in the training and validation sets respectively. The XGBoost model has the highest R^2 (0.88), the highest RMSE (16.554 μg/m^3), MAE (12.18 μg/m^3) and MAPE (29.673%). In comparison, Decision Trees has the lowest R^2 (0.71), the lowest RMSE (23.655 μg/m^3), MAE (17.733 μg/m^3) and MAPE (44.145%) models. Overall, the XGBoost model performs well and is able to predict O_3 mass concentrations.

Table 4 shows the results based on the VAR-tree model. Compared to the traditional tree model, we find a significant improvement for all 8 models. Fig. S2 shows the scatter plot of O_3 concentration estimated by the VAR-XGBoost, VAR-CatBoost, VAR-ExtraTrees, VAR-GBDT, VAR-AdaBoost, VAR-RF, VAR-Decision Tree and VAR-LightGBM models fits to the measured O_3 concentration at the ground monitoring sites. As it can be seen from Fig. 4, the VAR-tree model outperforms traditional machine learning models because the VAR-tree model is better at filtering features and modelling temporal variation, and it can better characterize the spatio-temporal characteristics of O_3. The fitted line rates of the scatter density maps drawn by the VAR-Adaboost model and VAR-XGBoost are 0.95 and 0.97 respectively, which indicates that the VAR-XGBoost model has the best fit, due to the introduction of the Kriging method in the VAR-XGBoost model, which fully takes into account the spatial correlation of the site data by calculating the variance function. Decision Trees fit is the least effective one, severely underestimating O_3 values and performs the worst. According to the value of MAPE, we can find that the maximum average error of CatBoost is more than 40%. Among the eight models, the values of MAPE of Adaboost, GBDT, XGBoost and LightGBM models are less than 20%, and the average error of VAR-XGBoost model is 17.266%. Comparing the eight machine learning tree models, it can be found that learningVAR-XGBoost model has the best prediction performance, followed by the VAR-Adaboost model, while VAR-CatBoost has the worst prediction performance among the eight models.

To test the accuracy of the VAR-XGBoost model simulations, a linear correlation analysis is performed between the simulated values of the O_3 model at the validation site and the actual measured values at the site. As shown in Fig. 4, the fitted R^2 for 2016, 2017, 2018, 2019, 2020 and 2021 are 0.93, 0.95, 0.95, 0.95, 0.95 and 0.95 respectively. The fitted R^2 for the 6-year mean value is 0.95, and the accuracy of the annual mean inversion is slightly higher than the accuracy of the quarterly mean inversion. The results show that the accuracy of the O_3 concentration in the inversion of the constructed VAR-XGBoost model is high. The training and validation results of the VAR-XGBoost model for 2016, 2017, 2018, 2019, 2020 and 2021 data show that the mean RMSE value is 11.831 μg/m^3, the mean MAE value is 7.686 μg/m^3, and the mean MAPE value is 17.67%. A density scatter plot of the fit results is shown in Fig. 4. There is a difference in model accuracy from 2016 to 2021. 2016 has the worst overall performance and 2019 has the best performance. Through validation, the VAR-XGBoost model is found to perform well on the whole with high accuracy.

Fig. S3 shows the scatter density plot of the fitted O_3 concentration estimated by the VAR-XGBoost model on a seasonal scale and the measured O_3 concentration at ground-based monitoring stations. A sample-based ten-fold cross-validation showed that the R^2 (0.94) was highest in spring. The highest RMSE (13.727 μg/m^3) and MAE (8.908 μg/m^3) were recorded highest during summer. The high correlation among mean temperature, mean precipitation and O_3 led to differences in R^2, which results in lower MAPE values during summer as well. Anthropogenic pollution leads to an increase in the O_3 mass concentrations and large estimation errors. In contrast, R^2 (0.91) was lowest in autumn, and winter was also the season with the lowest RMSE (7.188 μg/m^3) and MAE (4.741 μg/m^3). The lower estimation error in winter is due to the lower temperature in winter resulting in lower ground level O_3 mass concentrations and higher estimation accuracy. The higher MAPE (24.331%) in autumn is mainly due to the lower precipitation and lower temperatures in autumn compared to summer. Overall, the VAR-XGBoost model performed well on seasonal scales and was able to predict the distribution of O_3 mass concentrations on seasonal scales.

3.2. Spatial and temporal distribution characteristics of O_3 mass concentrations in China

3.2.1. Spatial and temporal seasonal distribution of O_3 concentrations in China

The seasons are first divided according to the climatic conditions of the Chinese region as a whole: spring from March to May, summer from June to August, autumn from September to November and winter from December to February. The spatial distribution of seasonal mean O_3 concentrations is shown in Fig. 5. The seasonal variation in ozone is also reflected spatially. In spring and summer, ozone pollution is worst in northern, eastern and central China. By autumn, the severest ozone pollution gradually shifted to the south. During the winter months, ozone pollution is relatively moderate across the country, with only a few cities in southern China experiencing ozone concentration exceeding the standard. Overall, there are few cities in China where ozone concentration is consistently good throughout the year, and ozone pollution is a more serious problem in the east than in the west. Among the four important urban agglomerations, the Beijing-Tianjin-Hebei

Table 4
Results of training and testing sets based on eight VAR-tree models.

	Statistic	Decision Trees	Random Forest	Adaboost	GBDT	CatBoost	Extra-tree	XGBoost	LightGBM
Training (80% of the data)	R^2	0.88	0.91	0.96	0.95	0.95	0.93	0.97	0.94
	RMSE	17.95	15.61	10.46	11.34	10.92	13.53	9.29	11.34
	MAE	11.83	11.11	6.79	7.34	7.98	11.48	5.99	7.46
	MAPE	25.18	21.35	14.74	12.76	13.33	23.27	9.05	17.42
Testing (20% of the data)	R^2	0.87	0.89	0.95	0.94	0.86	0.9	0.95	0.94
	RMSE	18.25	17.16	12.04	12.21	12.515	16.31	11.53	12.52
	MAE	12.93	12.21	7.55	8.07	8.44	12.49	7.475	8.363
	MAPE	30.01	28.41	17.3	18.83	22.815	27.43	17.26	19.54

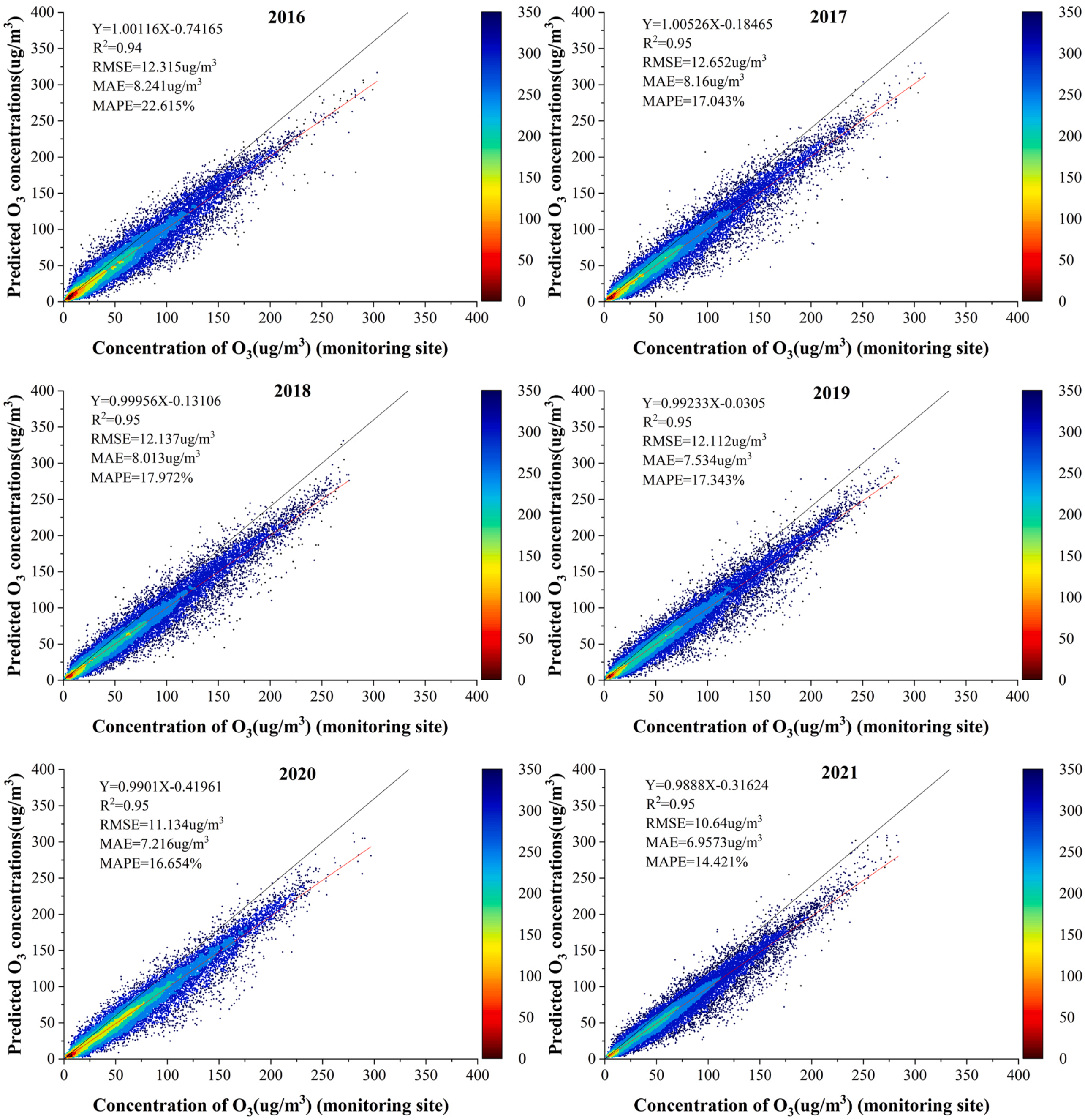

Fig. 4. O_3 simulation results for China 2016–2021 based on VAR-XGBoost.

urban agglomeration, the Yangtze River Delta urban agglomeration and the Chengdu-Chongqing urban agglomeration show large differences in seasonal variations in ozone concentration. In the Beijing-Tianjin-Hebei city clusters and the Yangtze River Delta city clusters, the highest ozone concentration tends to occur in summer. In the Chengdu-Chongqing urban agglomeration, there is little difference between spring and summer ozone concentration, which are both at high levels. In the PRD urban agglomeration, the highest ozone concentration occurs in autumn. In Chinese cities, there is an overall trend of high concentration in summer and low concentration in winter, and rising concentration in spring and falling concentration in autumn. The average O_3 concentration across cities in 2021 is 137 ug/m^3. Winter air quality is better in all cities, with pollution below 150 μg/m^3 in most areas. Beijing-Tianjin-Hebei region, East China, Central China and the Fenwei Plain are the most polluted area in summer, with most cities in the region exceeding 200 μg/m^3. In spring, the concentration is higher in the north than that in the south. In autumn, pollution is more severe in East China and Xinjiang. The extremely high value of seasonal pollution occurs in the summer in Xinjiang, reaching over 300 μg/m^3. The region suffers from some degree of pollution in all seasons except winter, which is relatively good. But Xinjiang's pollution in winter is still worst among all the cities which are with severe winter pollution across the country. It's summer pollution is at high level across the region. The overall O_3 quality in south China is good, and there is little difference in spring, autumn and winter, based on less than 150 μg/m^3. Winter pollution is relatively serious, which is mostly concentrated in Hunan and Jiangxi province. The seasonal distribution of O_3 concentrations in the Beijing-Tianjin-Hebei and Yangtze River Delta urban agglomerations is strongly influenced by temperature, with high temperatures in summer and strong solar radiation favoring photochemical reactions. While the difference

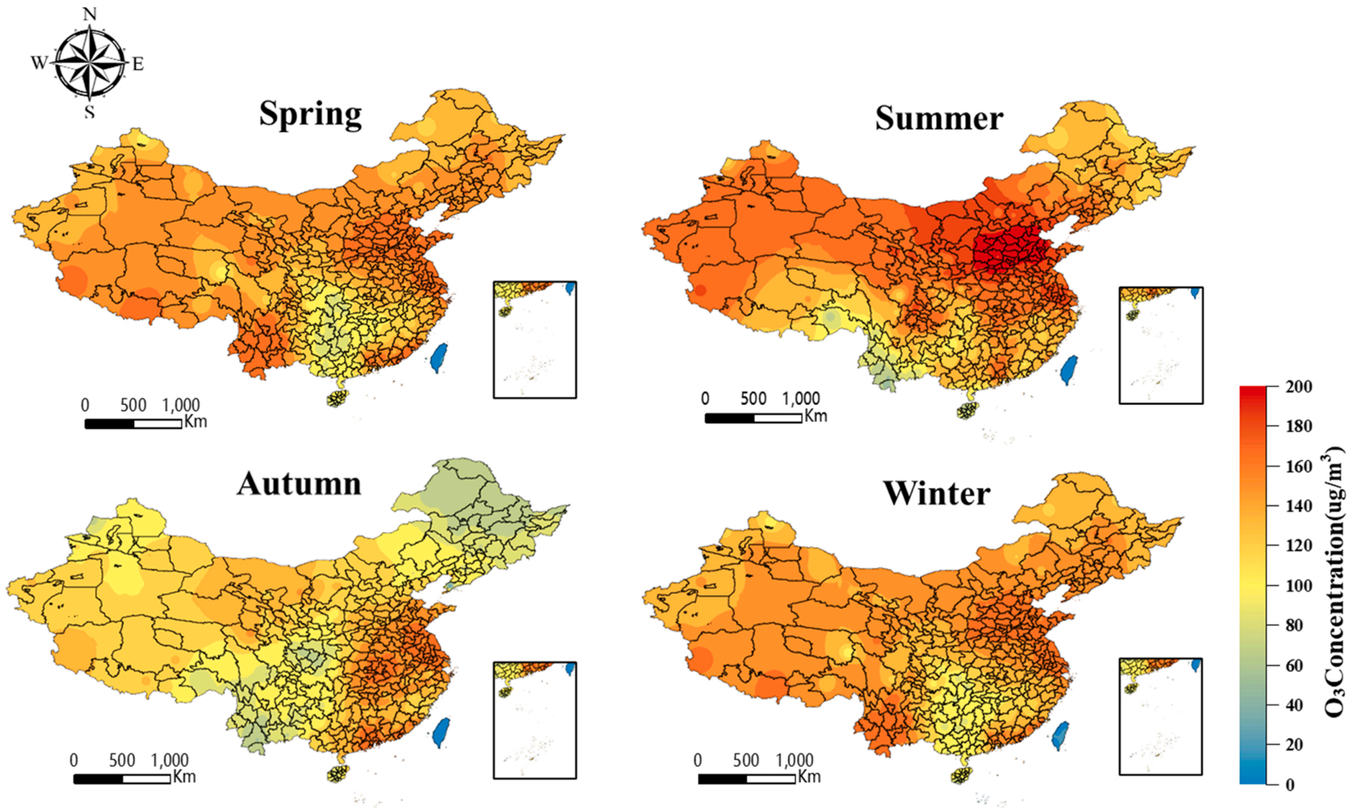

Fig. 5. Inversion of China's regional O_3 for the four seasons of 2021.

in temperature between spring and autumn is small. Intense solar radiation and higher temperatures in summer often lead to the production of photochemical smog and secondary ozone (Maji et al., 2019). Sustained high temperatures and intense sunny weather favour the photochemical reaction of atmospheric nitrogen oxides and volatile organic compounds, resulting in the production of surface ozone. Although summer temperatures are high in the PRD urban agglomeration, frequent precipitation inhibits the formation of O_3 and promotes its decomposition, and precipitation also has an impact on the inversion of satellite data.

3.2.2. Characteristics of interannual variability of O_3 concentrations in China

In this paper, a kriging interpolation analysis of O_3 concentration data for 361 cities in mainland China from 2016 to 2021 was conducted to obtain a six-year spatial distribution of O_3 concentrations. Fig. 6 shows a significant year-on-year growth trend in the total tropospheric O_3 column in all three major urban agglomerations from 2016 to 2021, with the Yangtze River Delta urban agglomeration growing the fastest and the Beijing-Tianjin-Hebei urban agglomeration growing the slowest. Of them, the Beijing-Tianjin-Hebei urban agglomeration changed steadily but maintained a high range after 2016, with the highest peak occurring in 2019. After a rapid increase in the Yangtze River Delta urban agglomeration after 2016 and peaks in 2018 and 2019, there was a sudden drop in total tropospheric O_3 columns in 2020, which was probably related to a reduction in O_3 precursor emissions during the COVID-19 pandemic. And a clear downward trend occurred in 2021, which was also influenced by the COVID-19 pandemic.

In terms of spatial aggregation distribution, in the northeast, ozone pollution gradually increases from north to south, and the cities with the most serious ozone pollution are mainly located in the coastal areas of Liaoning Province. In northern China, there is a trend that ozone pollution is high in the east and low in the west. The ozone pollution is relatively light in the Inner Mongolia Autonomous Region. In East China, the ozone concentration is higher in the northern cities than that in the southern cities. In south China, the areas with higher ozone concentrations are concentrated around the Pearl River Delta urban agglomeration. There is relatively less ozone pollution in the Guangxi Zhuang Autonomous Region and Hainan Province. In Central China, ozone concentrations gradually decrease from south to north, with Henan Province having the most prominent ozone pollution problem compared to the other two provinces. In the northwest, the highest ozone concentration is found near the city of Xi'an, and some cities in Gansu Province. And the Xinjiang Uyghur Autonomous Region also experienced ozone concentration exceeding the standard. In southwest China, ozone pollution is mainly concentrated around the Chengdu-Chongqing urban agglomeration, with relatively low levels of ozone concentrations in other areas.

As it can be seen from Fig. 6 that the overall pollution in North China has been severe for six years, with Beijing exceeding the standard for three consecutive years and being the most polluted city. Nine cities in Hebei Province have exceeded the standard for six consecutive years, with Tangshan, Baoding and Hengshui being the top three cities; central Inner Mongolia has a relatively lower level of pollution compared to central Inner Mongolia. Erdos has the highest pollution for six consecutive years. Compared to northern China, pollution is more concentrated in central China, which is mainly in Henan Province, with 14 cities of this region exceeding the standard. Among them, Jiaozuo is the most polluted one, exceeding the standard for six consecutive years. In the past six years, O_3 pollution in East China has become more and more serious. The pollution is obvious in Shanghai, Shandong Province and Jiangsu Province. There are up to 13 cities in Shandong province exceeding the standard, with Dezhou, Tai'an and Dongying being the top three cities. There are 10 cities in Jiangsu Province exceeding the standard, with Nanjing being the most polluted city. Overall pollution in South China is slight. The overall O_3 concentration in Guangdong

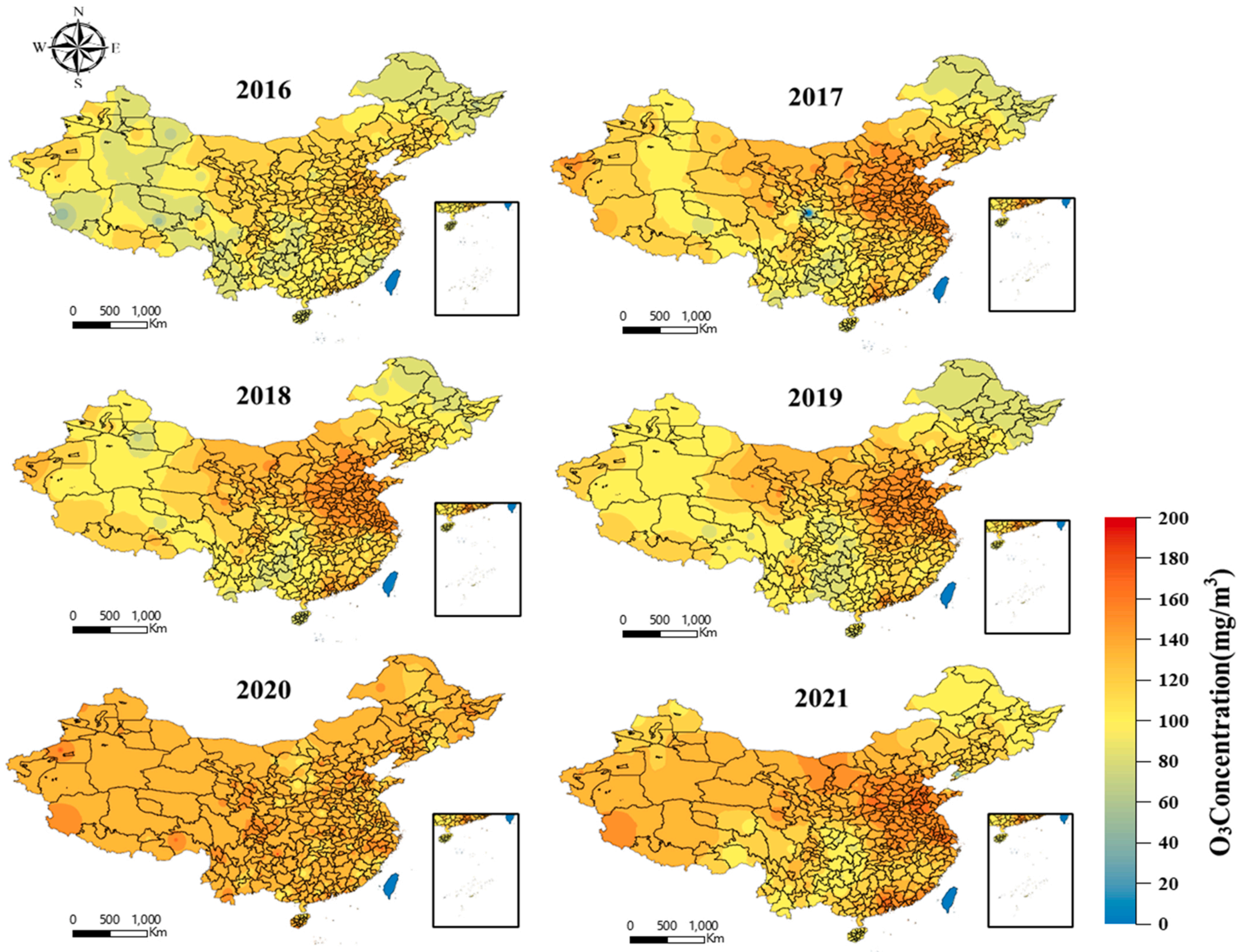

Fig. 6. Regional O_3 inversions in China, 2016–2021.

Province is high. But no cities have exceeded the standard. In the southwest, only some cities in Sichuan Province have exceeded the standard, with Chengdu being the most polluted one; in the northwest, the overall O_3 concentration is low. And no cities in the area have exceeded the standard. Pollution has been evident in Liaoning Province in the northeast, with four cities exceeding the standard for the sixth consecutive year, including the city of Yingkou, the most seriously-polluted city. The spatial distribution of the Kerrin interpolation of ozone concentrations from 2016 to 2021 shows that the regions with high ozone concentration are concentrated in Northwest, North, Central and East China. It also shows that Southwest and Northeast China have lower ozone concentrations than other regions due to a combination of factors such as altitude, topography and temperature. There are obvious regional differences in O_3 concentrations. From the perspective of inter-annual variation, the coastal areas of East China and South China in 2016 were the regions with high ozone concentrations.Over the following years(2016–2021), ozone concentrations in central and northwestern China gradually increased. Hebei Province, Gansu Province and Ningxia Hui Autonomous Region became the provinces with high ozone concentrations in China. Shandong Province, Henan Province and Jiangsu Province also had high ozone concentrations. Since 2017, a high-concentration area of ozone has formed in Northwest, North, Central and East China. This area follows a latitudinal distribution pattern, with high values being concentrated between 25° and 40°N. And the eastern coastal region has a higher concentration than inland regions. This spatial pattern is similar to the population distribution in China, probably due to the dense population, high level of urbanization and industrial development in the region. Therefore, more ozone precursors, such as volatile organic compounds and nitrogen oxides, are emitted during production and living processes. In northwest China, such as Gansu Province and Ningxia Hui Autonomous Region, long daylight hours and relatively high altitude are conducive to the chemical transformation of ozone precursors into ozone. As a result, a basic spatial pattern of ozone concentration distribution in China was formed, which has largely remained consistent in subsequent years. In terms of the stability of ozone concentration in each province, provinces such as Shandong, Jiangsu, Henan, Shanxi and Hebei have all steadily maintained a high level of ozone concentration, while provinces such as Heilongjiang and Guizhou have maintained a relatively low level of ozone concentration. Based on the above analysis and research, a combination of factors including dense population, high urbanization and long daylight hours lead to high ozone concentrations in some areas. In order to reduce the risk of human exposure and illness from ozone pollution, it is necessary to strengthen the regulation of ozone emissions in these highly-urbanized, industrialized and densely-populated areas to ensure a stable and sustained reduction in ozone pollution in the future.

3.2.3. Fitting effect of typical cities in China's O_3 heavily polluted regions

In China, the Beijing-Tianjin-Hebei region and its surrounding areas, the Yangtze River Delta and the Fenwei Plain are the regions with high emission intensity per unit area of air pollution sources. And these three regions are also the key areas identified by the state for air pollution prevention and control (Ministry of Ecology and Environment of the People's Republic of China, 2022). We fitted O_3 concentrations of 10 typical cities in heavily polluted areas. These cities are in the Beijing-Tianjin-Hebei region, the Fenwei Plain and the Yangtze River Delta, and they have large populations and a relatively large number of observation sites. As shown in Fig. S4, the R^2 for Shanghai is 0.88, and the R^2 for all nine cities, except Shanghai, is above 0.92. The values of RMSE and MAE show the highest accuracy in Beijing and Xi'an and worse results in Zhengzhou and Nanjing. We found that the accuracy in northern cities is higher than that in southern cities. And there are relatively large regional differences probably due to the lower winter temperatures and more snow and ice in northern cities, which are not conducive to O_3 formation. The higher temperatures in summer in the south affect the model's accuracy results.

4. Discussion

(1) Based on the training and testing datasets of ground station data and remotely sensed O_3 concentration, eight tree models as well as the VAR-tree model are established (VAR-XGBoost, VAR-CatBoost, VAR-ExtraTrees, VAR-GBDT, VAR-AdaBoost, VAR-RF, Decision Tree, VAR- LightGBM model). The VAR-XGBoost model outperforms the other tree models, with a coefficient of determination R^2 around 0.9 for both training and prediction accuracy. The prediction accuracy and reliability of VAR-XGBoost are better. In order to verify that the O_3 concentration prediction, which is based on the VAR-XGBoost model, is more accurate, the validation is carried out from the perspective of different datasets and different control models. From the horizontal datasets, we find that the XGBoost model has the highest prediction accuracy under the sample datasets by predicting XGBoost, CatBoost, ExtraTrees, GBDT, AdaBoost, RF, Decision Tree, LightGBM, where the VAR-tree model-based prediction of O_3 concentration results in a 12.69% increase in R^2, a 24.28% decrease in RMSE, a 29.69% decrease in MAE and a 34.68% decrease in MAPE compared to the eight-tree model-based prediction of O_3 concentration. From the longitudinal datasets, the VAR-XGBoost model has a higher prediction accuracy over VAR-CatBoost, VAR-ExtraTrees, VAR-GBDT, VAR-AdaBoost, VAR-RF, VAR-Decision Tree, and VAR-LightGBM, where the VAR-XGBoost model has an average 42.63% increase in R^2, a 36.83% decrease in RMSE, a 42.19% decrease in MAE and a 42.48% decrease in MAPE compared to the VAR-Decision Trees model. Compared to the VAR-Random Forest model, the VAR-XGBoost model has a 34.23% decrease in RMSE, s 42.37% decrease in MAE and a 42.48% decrease in MAPE. Compared to the VAR-Adaboost model, VAR-XGBoost model has a 4.24% decrease in RMSE, 0.99% decrease in MAE and a 0.2% decrease in MAPE. Compared to the VAR-GBDT model, the VAR-XGBoost model has a 5.56% decrease in RMSE, a 7.37% decrease in MAE, and a 8.29% decrease in MAPE. Compared to the VAR-CatBoost model, the VAR-XGBoost model has a 28.77% decrease in RMSE, a 38.25% decrease in MAE, and a 58.31% decrease in MAPE 58.31%. Compared to the VAR-Extra-tree model, the VAR-XGBoost model reduced RMSE by 29.3%, MAE by 40.16%, and MAPE by 37.06%. Compared to the VAR-LightGBM model, the LUR-GBM model has a 7.88% decrease in RMSE, a 10.62% decrease in MAE and a 11.65% decrease in MAPE.

(2) We find that the effects of PM_{10}, SO_2, NO_2 and CO concentrations on O_3 concentration are long-lasting and positive, with increases in NO_2 and CO concentrations causing increases in O_3 concentration through the VAR variance decomposition. However, the overall effect of O_3 concentration on $PM_{2.5}$ concentration is negative, with an increase in O_3 concentration causing a decrease in $PM_{2.5}$ concentration. The variance decomposition shows that, among these variables, NO_2 and CO concentrations have the greatest influence on O_3 concentration with a strong lag effect, and O_3 has a suppressive effect on $PM_{2.5}$ concentration.

(3) Based on the VAR-XGBoost prediction model and the spatial and temporal distribution characteristics of O_3 in various regions of China, this paper finds that the annual average O_3 concentration in various regions of China increased from 2016 to 2021. Among them, the Beijing-Tianjin-Hebei region has experienced the fastest rise and the most serious pollution due to economic growth and regional transport of ozone precursors. Due to the impact of the epidemic in 2020, the concentration of O_3 decreased slightly. The spatial and temporal distribution of surface O_3 concentration in China reveals that the distribution of O_3 concentration in China is characterized by "high in summer and low in winter, rising in spring and falling in autumn". In particular, O_3 pollution is most severe in winter in areas such as central China. In spring, O_3 pollution is most severe in Eastern China and Xinjiang. In autumn, O_3 concentration is higher in Northern China than that in Southern China (Chen et al., 2022a). In summer, O_3 pollution is severe in the Beijing-Tianjin-Hebei region (Wang et al., 2022b). In terms of the spatial distribution of O_3 concentration, China's pollution regions are characterized by "higher levels in the east than in the west".

(4) Exploring the distribution characteristics of ozone concentration across the country can help provide a theoretical basis and data support for air pollution prevention and control and environmental protection policies, etc. The national average O_3 concentration decreased slightly from 138 ug/m^3 to 137 ug/m^3 between 2016 and 2021. Ozone concentration in Chinese cities exceed standards, with almost no cities meeting the annual limit for primary ozone concentration, and nearly a third failing to meet the limit for secondary concentration. Within regions and urban agglomerations, core cities with large populations, developed economies and industries tend to have higher ozone concentration levels. The most serious pollution is found in northern China, mainly including Southern Hebei, Northern Henan and Western Shandong. This is followed by greater ozone pollution in central China, the Sichuan basin and southern China. The lowest O_3 concentrations is found in Yunnan and Northeast China. Heavily polluted areas are in the Beijing-Tianjin-Hebei region, the Fenwei Plain, Eastern China and Western Xinjiang. The overall pollution situation remains grim, with O_3 concentration in the Beijing-Tianjin-Hebei region and the Fenwei Plain still being high. In general, ozone pollution is serious, with O_3 concentration in the Beijing-Tianjin-Hebei region and the Fenwei Plain is high, and the overall pollution area of the country is large. In the Beijing-Tianjin-Hebei urban agglomeration, ozone pollution is characterized by regional clustering, with a tendency to expand outward from the center (Xu et al., 2022).

(5) We studied and discussed O_3 concentration predictions in ten typical cities in heavily polluted areas in China, and then we found that the O_3 concentration forecast accuracy of northern cities is higher than that of southern cities. Among them, Beijing and Tianjin have the highest degree of forecast accuracy, while Shanghai and Nanjing have a lower degree of forecast accuracy.

(6) Temperature, sunshine hours and air pressure are significant factors influencing O_3 concentration near the ground. Higher temperatures, longer sunshine hours and lower air pressure will lead to a higher O_3 concentration (Bi et al., 2022). High relative humidity has a significant suppressive effect on O_3 concentration in summer, but it has no significant effect during the rest of the season; the relationship between O_3 and wind speed is more complex (Lee et al., 2023). High $PM_{2.5}$ concentration can absorb and scatter atmospheric UV radiation and reduce atmospheric O_3 level (Duan et al., 2022). The relationship between NO_2 and O_3 is mainly related to the season. In summer, NO_2 reacts with other precursors to produce O_3. However, in winter, the photochemical reaction is limited, and ozone production is weakened, so NO_2 continues to increase (Xing et al., 2022).

5. Conclusion

In this paper, a typical hybrid model, the VAR-tree model, is proposed to estimate the spatial and temporal distribution of O_3 concentration by using atmospheric pollutant data and conventional meteorological observation elements (e.g. air temperature, surface temperature, wind speed, wind direction, relative humidity, pressure and visibility) based on O_3 observation data in China from 2016 to 2021. By analyzing the spatial and temporal patterns of O_3 and its influencing factors, this paper clarifies the changes in O_3 at different time scales and the underlying mechanisms in recent years. Finally, the general patterns of spatial and temporal distribution of O_3 concentrations in China are summarized. Therefore, through inversion of O_3, it helps to grasp the regional variation process of O_3 in time and space. This paper provides a scientific basis for regional O_3 pollution prevention and control, and provides new ideas for management departments to obtain data on the spatial distribution of O_3 concentrations. The VAR-XGBoost method can better solve the problem of spatial heterogeneity of the study object.

CRediT authorship contribution statement

Hongbin Dai: Conceptualization, Methodology, Writing – original draft. **Guangqiu Huang:** Conceptualization, Writing – review & editing. **Jingjing Wang**: editing, revision. **Huibin Zeng:** Writing – review & editing.

Funding

This paper is funded by the National Natural Science Foundation of China (71874134). Key Project of Basic Natural Science Research Plan of Shaanxi Province (2019JZ-30). Key research projects of Scientific Research Fundation of Guangxi Science and Technology Normal University (GXKS2022ZD001)

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2023.114960.

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